# Artificial Intelligence

# Course Outcomes

| COURSE OUTCOMES | | COGNITIVE LEVELS |
|---|---|---|
| 1 | Understand the basics of artificial intelligence, problem solving strategies, and machine learning methods | Understand Level(C2) |
| 2 | Apply intelligent searching techniques to solve a given problem | Apply Level (C3) |
| 3 | Apply supervised, unsupervised or semi-supervised learning algorithms for a given problem | Apply Level (C3) |
| 4 | Apply the machine learning algorithms for classification problems | Apply Level (C3) |
| 5 | Analyze the different deep learning architectures for real-world applications. | Analyze Level(C4) |

# Text book

- **Rich and Knight,** “Artificial Intelligence”, Tata  McGraw Hill, 1992
- **S. Russel and P. Norvig,** “Artificial Intelligence – A Modern Approach”, Second Edition, Pearson  Edu.

# Definition of AI

- AI is one of the fascinating and universal fields of Computer science which has a great scope in future.
- AI holds a tendency to cause a machine to work as a human.
- Artificial Intelligence is composed of two words **Artificial** and **Intelligence**, where Artificial defines *"man-made,"* and intelligence defines *"thinking power"*, hence AI means *"a man-made thinking power.“*
- Replicate human intelligence
- Solve Knowledge-intensive tasks

# Definition of Artificial Intelligence

"It is a branch of computer science by which we can create intelligent machines which can behave like a human, think like humans, and able to make decisions."

General Definition

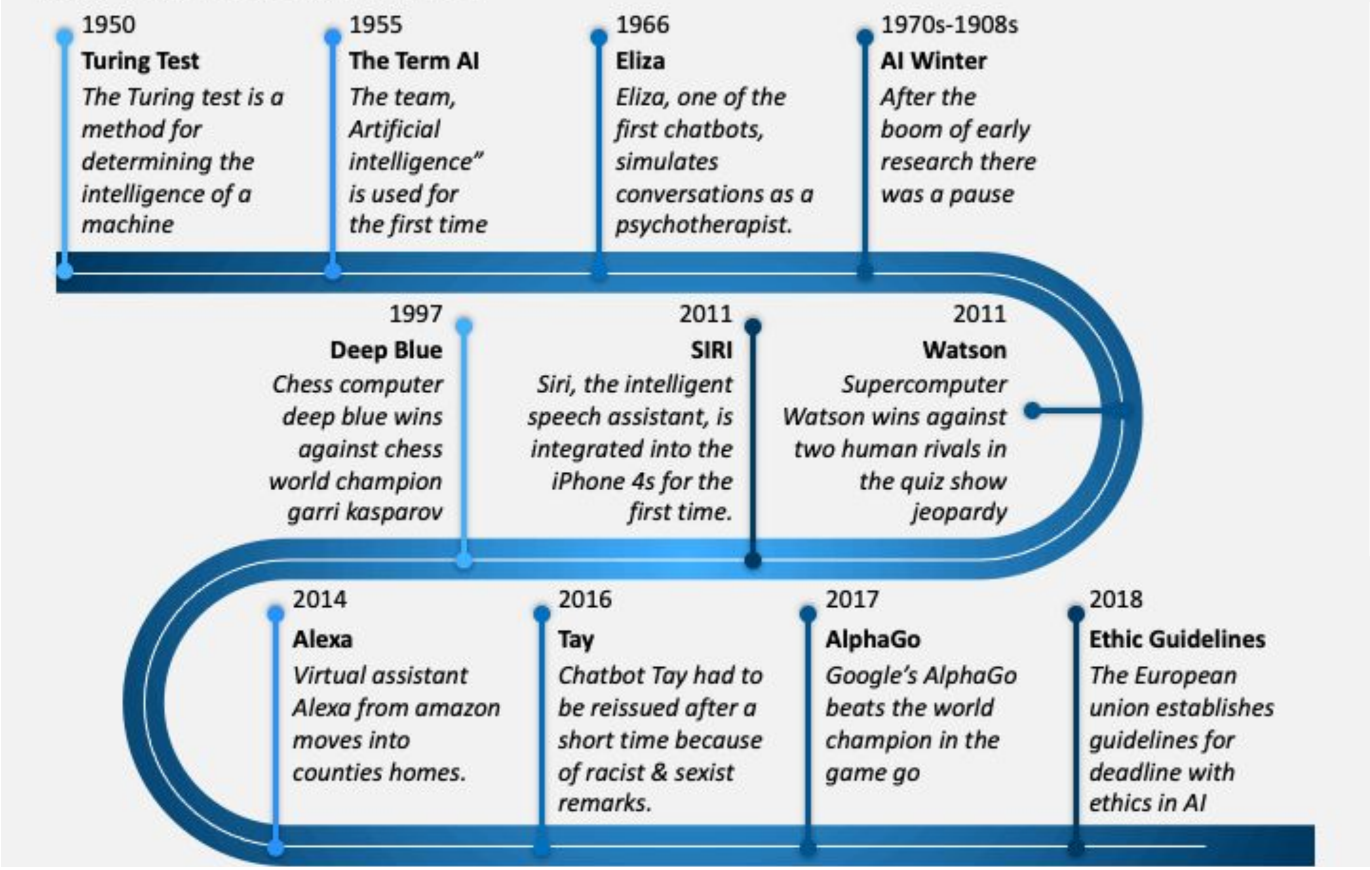
"The exciting new effort to make computers think.. Machine with minds, in the full literal sense "

Haugeland, 1985

"The study of how to make computers do things at which, at the moment, people are better."

Rich & Knight, 1991

# History of AI




- **The gestation of artificial intelligence (1943-1955)**
- **The birth of artificial intelligence (1956)**
- **Early enthusiasm, great expectations (1952-1969)**
- **A dose of reality (1966-1973)**
- **Knowledge-based systems: The key to power? (1969-1979)**
- **AI becomes an industry (1980-present)**
- **The return of neural networks (1986-present)**
- **AI becomes a science (1987-present)**
- **The emergence of intelligent agents (1995-present)**

# Motivation of Artificial Intelligence

- Computers are fundamentally well suited to performing mechanical computations, using fixed programmed rules.

- Artificial machines perform simple monotonous tasks efficiently and reliably, which humans are ill-suited to.

- For more complex problems, things get more difficult... Unlike humans, computers have trouble understanding specific situations, and adapting to new situations.

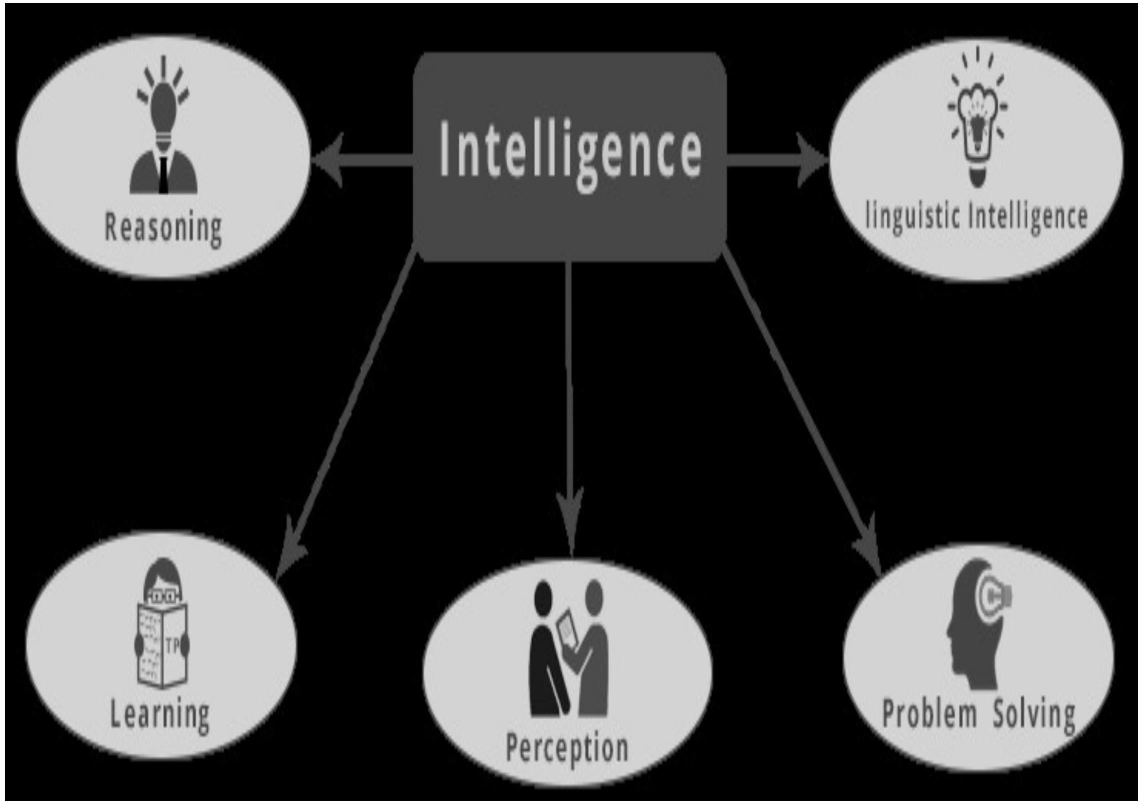
- Artificial Intelligence aims to improve machine behavior in tackling such complex tasks.

# What is Intelligence?

- Ability of a system to calculate, reason, perceive relationship, and analogies ,store, retrieve information from memory , solve problems, etc.
  - take decision
  - Assumptions
  - classify
  - use of knowledge to respond new situations
  - ability to learn
  - Inductive inference

# Type of Intelligence:

- Linguistic Intelligence
- Musical Intelligence
- Logical Mathematical Intelligence
- Spatial Intelligence
- Bodily – Kinesthetic Intelligence
- Intra Personal Intelligence
- Interpersonal intelligence



# Characteristics of AI systems

- learn new concepts and tasks
- reason and draw useful conclusions about the world around us
  - remember complicated interrelated facts and draw conclusions from them (inference)
- understand a natural language or perceive and comprehend a visual scene
  - look through cameras and see what's there (vision), to move themselves and objects around in the real world (robotics)

# Contd..

- plan sequences of actions to complete a goal
- offer advice based on rules and situations
- may not necessarily imitate human senses and thought processes
  - but indeed, in performing some tasks differently, they may actually exceed human abilities
- capable of performing intelligent tasks effectively and efficiently
- perform tasks that require high levels of intelligence

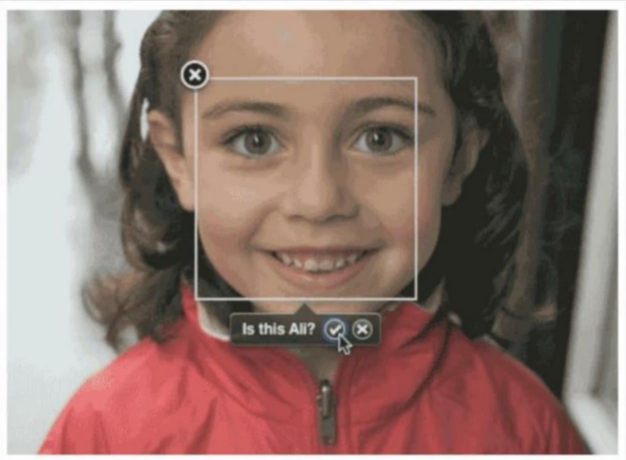
# Research Areas of AI

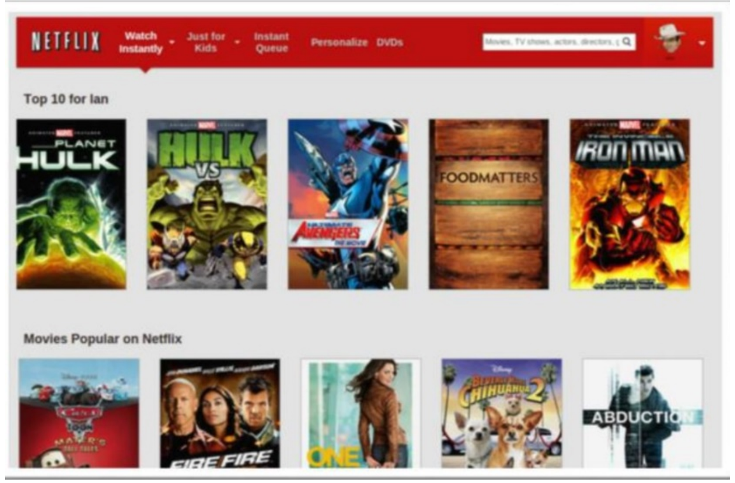
- Problem Solving through heuristic techniques
- Knowledge representation
- Handling uncertain situations
- Theorem proving
- Game playing
- Natural language processing
- Expert system

# Subareas of AI

- Perception: vision, speech understanding, etc.
- Machine Learning, Neural networks
- Robotics
- Natural language processing
- Reasoning and decision making
  - Knowledge representation
  - Reasoning (logical, probabilistic)
  - Decision making (search, planning, decision theory)

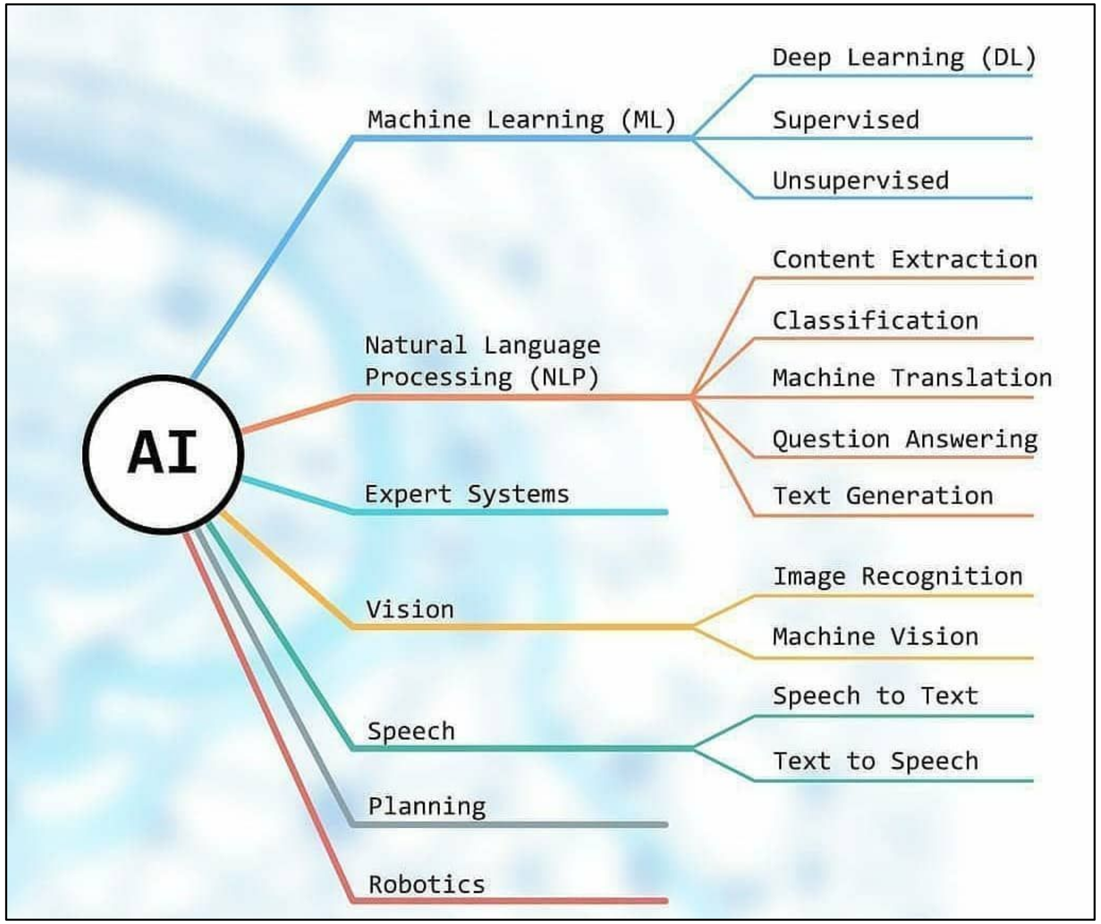
# Applications of AI

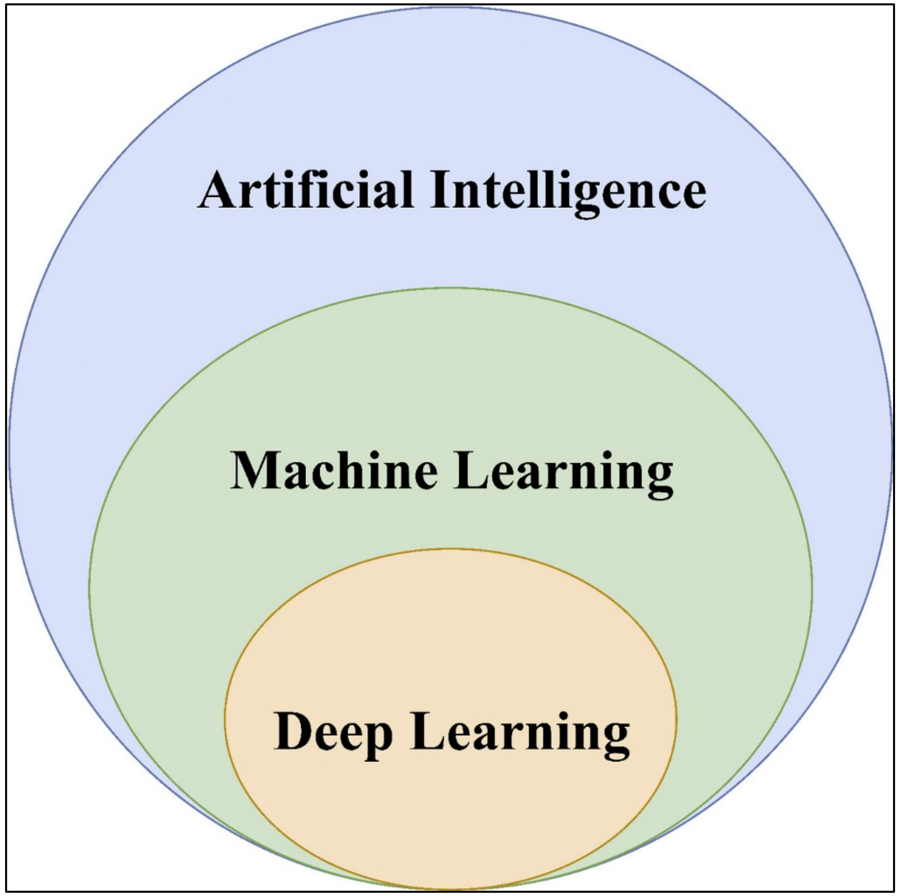




- Chatbots
- Astronomy
- Gaming
- Finance
- Data Security
- Travel and Transport
- Marketing
- Entertainment
- Military
- Healthcare
- GPS and Navigations
- Automobiles
- Agriculture
- Human Resource
- Lifestyle

# Artificial Intelligence Techniques



# Advantage of AI

Following are some main advantages of Artificial Intelligence:

- **High Accuracy with less errors**: AI machines or systems are prone to less errors and high accuracy as it takes decisions as per pre-experience or information.
- **High-Speed**: AI systems can be of very high-speed and fast-decision making, because of that AI systems can beat a chess champion in the Chess game.
- **High reliability**: AI machines are highly reliable and can perform the same action multiple times with high accuracy.
- **Useful for risky areas**: AI machines can be helpful in situations such as defusing a bomb, exploring the ocean floor, where to employ a human can be risky.
- **Digital Assistant**: AI can be very useful to provide digital assistant to the users such as AI technology is currently used by various E-commerce websites to show the products as per customer requirement.
- **Useful as a public utility**: AI can be very useful for public utilities such as a self-driving car which can make our journey safer and hassle-free, facial recognition for security purpose, Natural language processing to communicate with the human in human-language, etc.

# Disadvantage of AI

Every technology has some disadvantages, and the same goes for Artificial intelligence. Being so advantageous technology still, it has some disadvantages which we need to keep in our mind while creating an AI system. Following are the disadvantages of AI:

- **High Cost:** The hardware and software requirement of AI is very costly as it requires lots of maintenance to meet current world requirements.
- **Can't think out of the box:** Even we are making smarter machines with AI, but still they cannot work out of the box, as the robot will only do that work for which they are trained, or programmed.
- **No feelings and emotions:** AI machines can be an outstanding performer, but still it does not have the feeling so it cannot make any kind of emotional attachment with human, and may sometime be harmful for users if the proper care is not taken.
- **Increase dependency on machines:** With the increment of technology, people are getting more dependent on devices and hence they are losing their mental capabilities.
- **No Original Creativity:** As humans are so creative and can imagine some new ideas but still AI machines cannot beat this power of human intelligence and cannot be creative and imaginative.

# Categories of AI System

- Systems that think like humans
- Systems that act like humans
- Systems that think rationally
- Systems that act rationally

# Systems that think like humans

- Most of the time it is a black box where we are not clear about our thought process.
- One has to know functioning of brain and its mechanism for possessing information.
- It is an area of **cognitive science**.
  - The stimuli are converted into mental representation.
  - Cognitive processes manipulate representation to build new representations that are used to generate actions.
- Neural network is a computing model for processing information similar to brain.

# Systems that act like humans

- The overall behavior of the system should be human like.
- It could be achieved by observation.

# Systems that think rationally

- Such systems rely on logic rather than human to measure correctness.
- For thinking rationally or logically, logic formulas and theories are used for synthesizing outcomes.
- For example,

  Eg: John is a human and all humans are mortal then one can conclude logically that John is mortal
- Not all intelligent behavior are mediated by logical deliberation.

# Systems that act rationally

- Rational behavior means doing right thing.
- Even if method is illogical, the observed  behavior must be rational.

# Approaches to AI

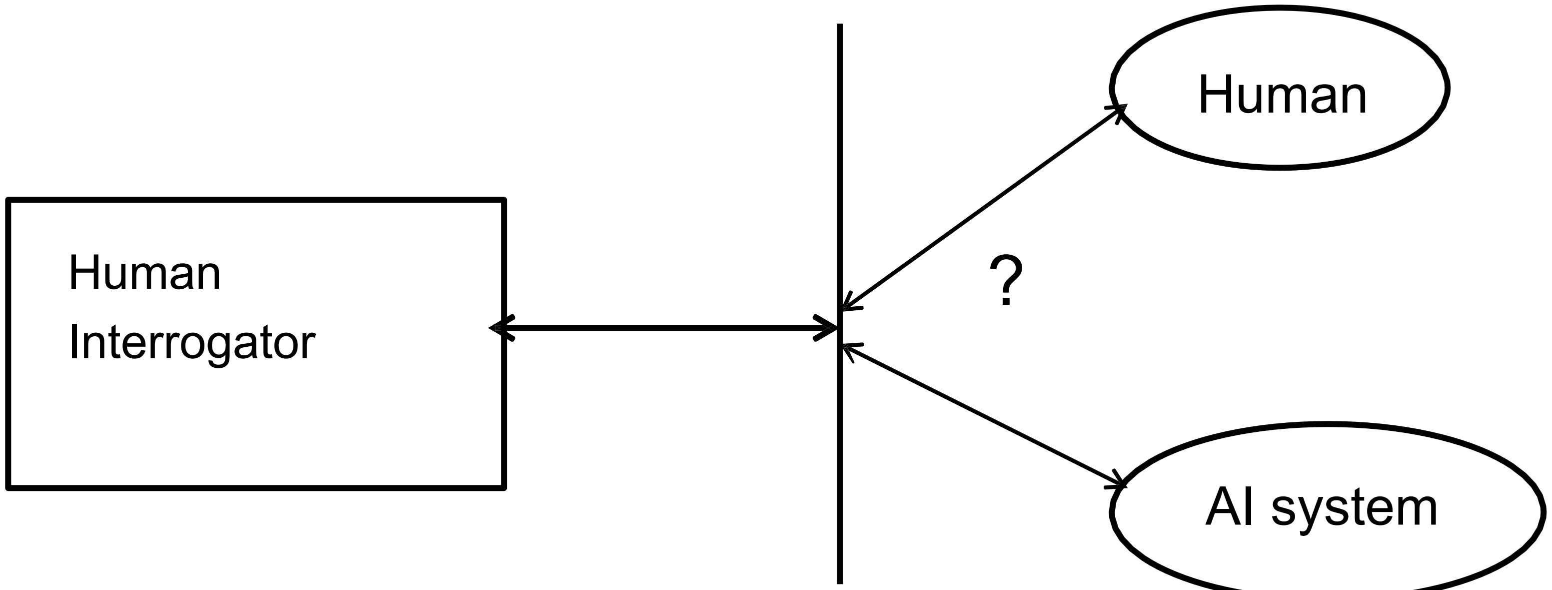
- Strong AI aims to build machines that can truly reason and solve problems. These machines should be self aware and their overall intellectual ability needs to be indistinguishable from that of a human being. Excessive optimism in the 1950s and 1960s concerning strong AI has given way to an appreciation of the extreme difficulty of the problem. Strong AI maintains that suitably programmed machines are capable of cognitive mental states.
- Weak AI: deals with the creation of some form of computer-based artificial intelligence that cannot truly reason and solve problems, but can act as if it were intelligent. Weak AI holds that suitably programmed machines can simulate human cognition.
- Applied AI: aims to produce commercially viable "smart" systems such as, for example, a security system that is able to recognise the faces of people who are permitted to enter a particular building. Applied AI has already enjoyed considerable success.
- Cognitive AI: computers are used to test theories about how the human mind works--for example, theories about how we recognise faces and other objects, or about how we solve abstract problems.

# The Turing Test: Preliminaries

- Designed by Alan Turing (1950)
- The Turing test provides a satisfactory **operational** definition  of AI
- It's a behavioral test (i.e., test if a system acts like a human)
- Problem: it is difficult to make a mathematical analysis of it.

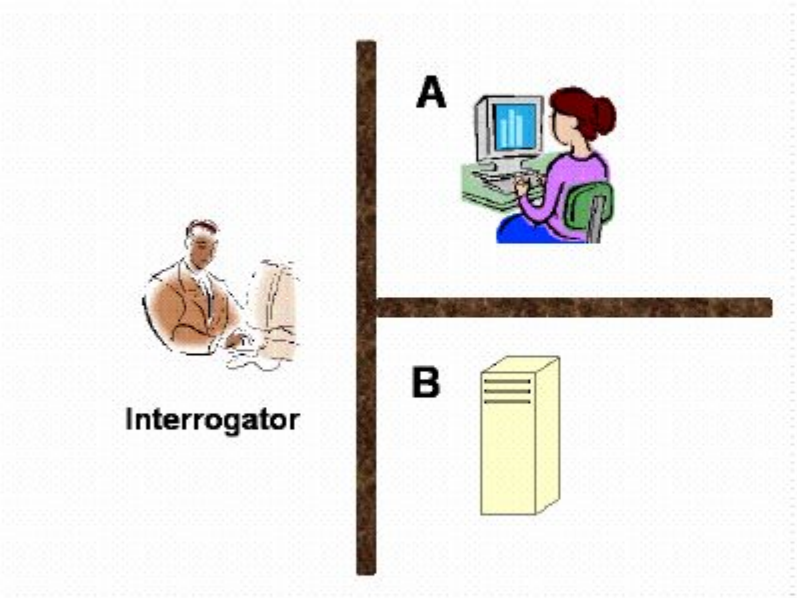
# The Turing Test

Turing proposed operational test for intelligent  behavior in 1950.



# The Turing Test

- There are two rooms, A and B. One of the rooms contains a computer. The other contains a human. The interrogator is outside and does not know which one is a computer. He can ask questions through a teletype and receive answers from both A and B. The interrogator needs to identify whether A or B are humans. To pass the Turing test, the machine has to fool the interrogator into believing that it is human.



# Typical AI problems

While studying the typical range of tasks that we might expect an “intelligent entity” to perform, we need to consider both “common-place” tasks as well as expert tasks.

Examples of common-place tasks include

– *Recognizing* people, objects.

– Communicating (through *natural language*).

– *Navigating* around obstacles on the streets

These tasks are done matter of factly and routinely by people and some other animals

# Typical AI problems – Cont.

Expert tasks include:

- Medical diagnosis.
- Mathematical problem solving
- Playing games like chess

These tasks cannot be done by all people, and can only be performed by skilled specialists.

Now, which of these tasks are easy and which ones are hard?

# Typical AI problems – Cont.

- Clearly tasks of the first type are easy for humans to perform, and almost all are able to master them.
- The second range of tasks requires skill development and/or intelligence and only some specialists can perform them well.
- However, when we look at what computer systems have been able to achieve to date, we see that their achievements include performing sophisticated tasks like medical diagnosis, performing symbolic integration, proving theorems and playing chess.

# Problem Solving

Problem solving is a process of generating solutions from observed data.

- a **Problem** is characterized by a set of goals,
- a set of **objects**, and
- a set of **operations**.

# Problem

**Problem definitions**

A problem is defined by its '*elements*' and their '*relations*'. To provide a formal description of a problem, we need to do the following:

a. Define a *state space* that contains all the possible configurations of the relevant objects, including some impossible ones.
b. Specify one or more states that describe possible situations, from which the problem-solving process may start. These states are called *initial states*.
c. Specify one or more states that would be acceptable solution to the problem.

These states are called *goal states*.

# Problem Solving –cont.

To solve the problem of building a system you should take the following steps:

1. Define the problem accurately including detailed specifications and what constitutes a suitable solution.
2. Scrutinize the problem carefully, for some features may have a central affect on the chosen method of solution.
3. Segregate and represent the background knowledge needed in the solution of the problem.
4. Choose the best solving techniques for the problem to solve a solution.

# Steps require to solve a problem

These are the following steps which require to solve a problem :

- **Problem definition:** Detailed specification of inputs and acceptable system solutions.
- **Problem analysis:** Analyse the problem thoroughly.
- **Knowledge Representation:** collect detailed information about the problem and define all possible techniques.
- **Problem-solving:** Selection of best techniques.

# Problem Solving –cont.

- **Problem solving:** The term, Problem Solving relates to analysis in AI. Problem solving may be characterized as a systematic search through a range of possible actions to reach some predefined goal or solution. Problem-solving methods are categorized as *special purpose* and *general purpose*.

- A *special-purpose method* is tailor-made for a particular problem, often exploits very specific features of the situation in which the problem is embedded.

- A *general-purpose method* is applicable to a wide variety of problems. One General-purpose technique used in AI is *'means-end analysis'*: a step-bystep, or incremental, reduction of the difference between current state and final goal.

# Components to formulate the associated problem

- **Initial State:** This state requires an initial state for the problem which starts the AI agent towards a specified goal. In this state new methods also initialize problem domain solving by a specific class.
- **Action:** This stage of problem formulation works with function with a specific class taken from the initial state and all possible actions done in this stage.
- **Transition:** This stage of problem formulation integrates the actual action done by the previous action stage and collects the final stage to forward it to their next stage.
- **Goal test:** This stage determines that the specified goal achieved by the integrated transition model or not, whenever the goal achieves stop the action and forward into the next stage to determines the cost to achieve the goal.
- **Path costing:** This component of problem-solving numerical assigned what will be the cost to achieve the goal. It requires all hardware software and human working cost.

# Core Components of Artificial Intelligence

Intelligent systems combine perception, learning, reasoning, planning, and search to act autonomously and effectively in complex environments.

- **Perception** → understand environment
- **Learning** → improve from data
- **Reasoning** → draw conclusions
- **Planning** → decide actions
- **Search** → find optimal solutions

# Search

- Explores possible states to find a solution
- Uses **state space**, **initial state**, **goal state**
- Types:
  - Uninformed: BFS, DFS, UCS
  - Informed: A*, Greedy Best-First
- Uses **heuristic functions** to guide search
- Key concerns: **completeness, optimality, time & space complexity**
- Applications: pathfinding, game playing, scheduling

# Reasoning

- Uses **logic and inference** to derive conclusions
- Based on **knowledge representation**
- Types:
  - Deductive (rule-based)
  - Inductive (learning from examples)
  - Abductive (best explanation)
- Handles uncertainty using **probability** and **fuzzy logic**
- Applications: expert systems, diagnosis, decision support

# Planning

- Determines **sequence of actions** to reach goals
- Uses:
  - Preconditions
  - Effects
  - Action costs
- Types:
  - Classical planning
  - Temporal planning
  - Probabilistic planning
- Often uses **search + reasoning**
- Applications: robotics, logistics, autonomous agents

# Learning

- Enables systems to **improve with experience**
- Learns patterns from **data or feedback**
- Types:
  - Supervised learning
  - Unsupervised learning
  - Reinforcement learning
- Models: decision trees, neural networks
- Applications: prediction, recommendation, classification

# Perception

- Interprets sensory input from the environment
- Converts raw data into **high-level features**
- Modalities:
  - Vision (images, video)
  - Speech & audio
  - Sensors (IoT, robotics)
- Challenges: noise, ambiguity, incomplete data
- Applications: face recognition, speech recognition, autonomous driving

# Tic-Tac-Toe

X O

e(p) = 6 - 5 = 1

- **Initial State**: Board position of 3x3 matrix with 0 and X.
- **Operators:** Putting 0's or X's in vacant positions alternatively
- **Terminal test:** Which determines game is over
- **Utility function**:

  e(p) = (No. of complete rows, columns or diagonals are still open for player ) – (No. of complete rows, columns or diagonals are still open for opponent )

# Minimax Algorithm

Generate the game tree

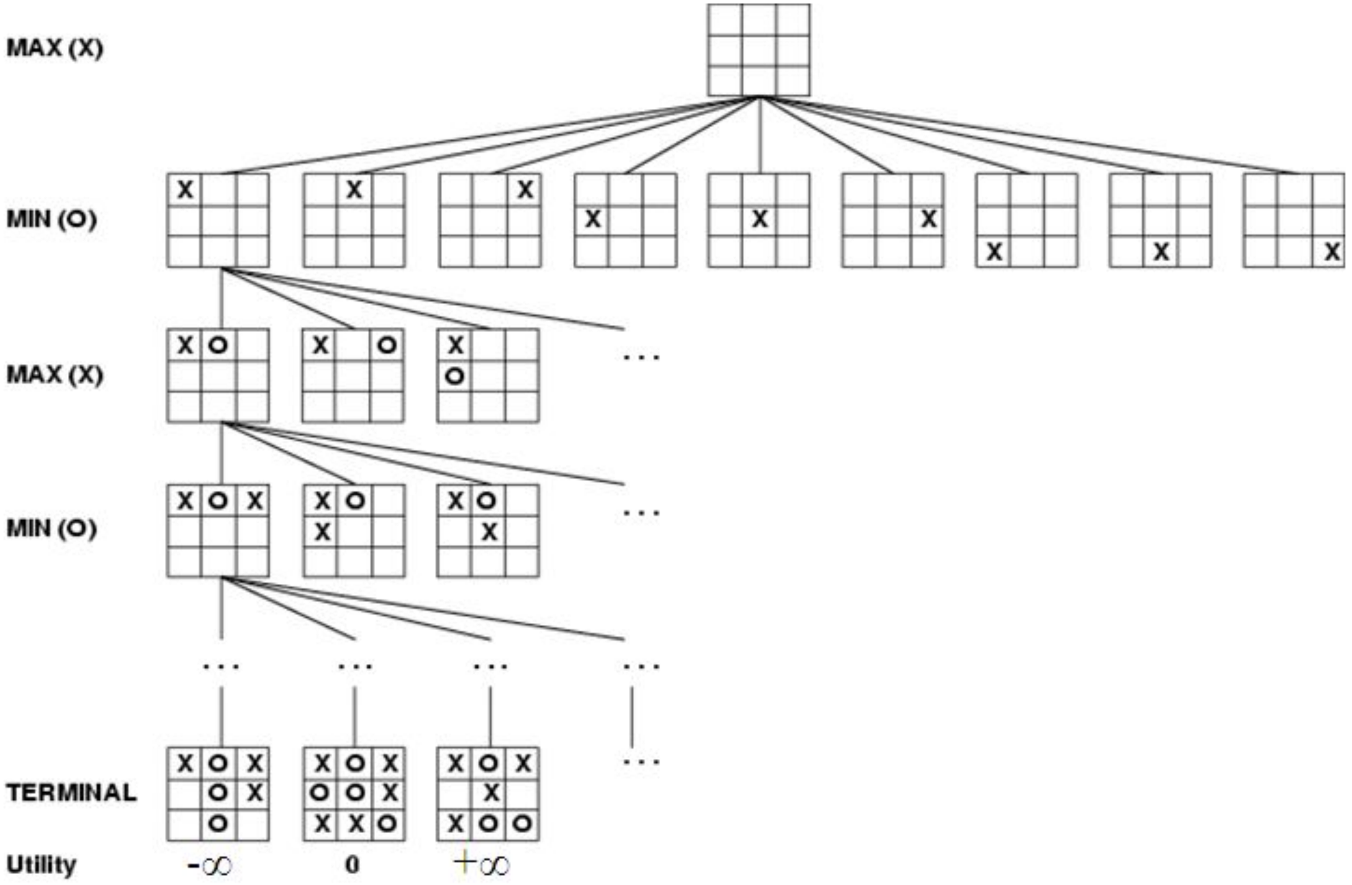
Apply the utility function to each terminal state to get its value

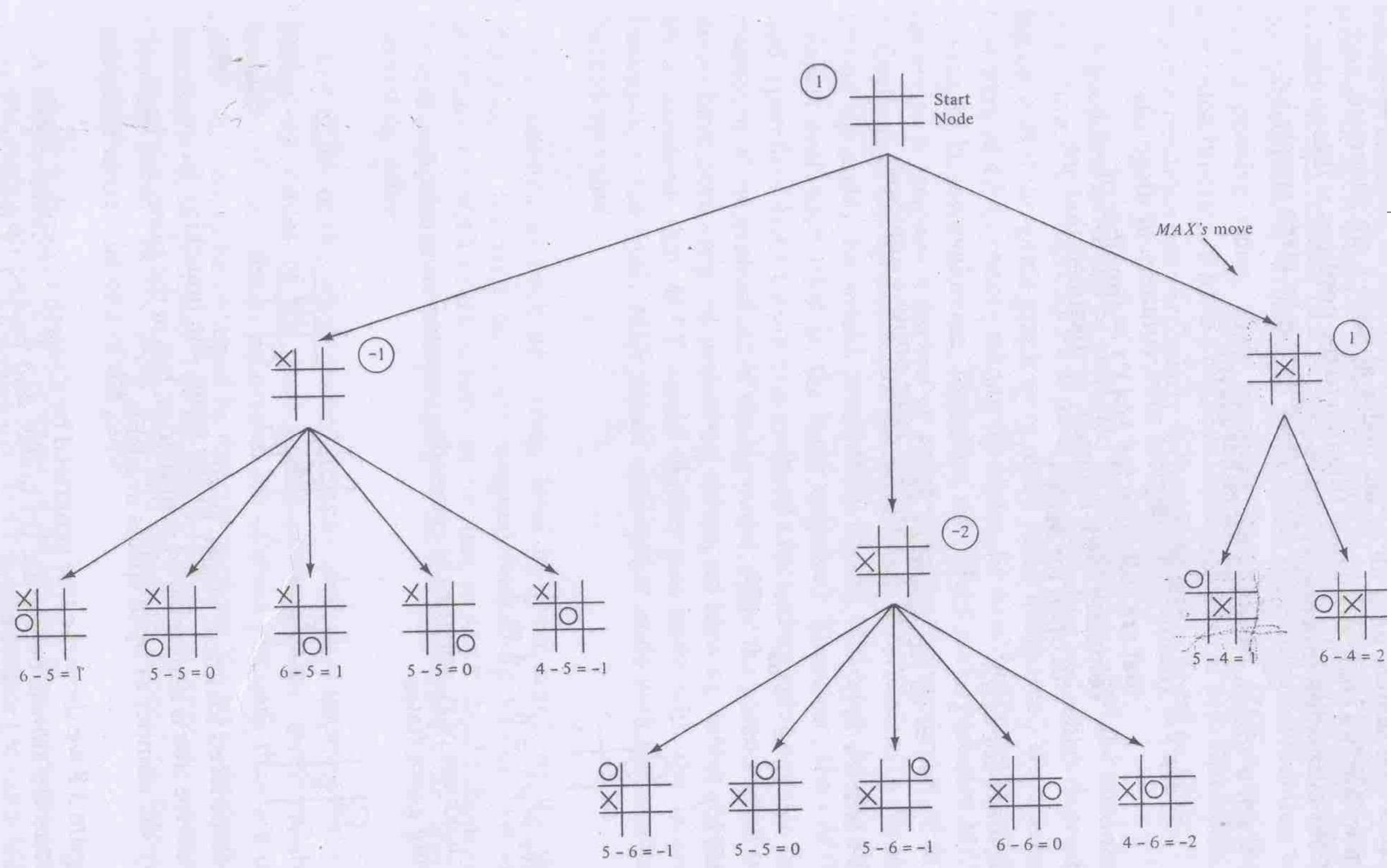
Use these values to determine the utility of the nodes one level higher up in the search tree

- From bottom to top
- For a max level, select the maximum value of its successors
- For a min level, select the minimum value of its successors

From root node select the move which leads to highest value

# Game tree for Tic-Tac-Toe






Courtesy : Principles of Artificial Intelligence , Nilsson

# Properties of Minimax

Complete : Yes (if tree is finite)

Time complexity : $O(b^d)$

Space complexity : $O(bd)$ (depth-first exploration)

# Observation

Minimax algorithm, presented above, requires expanding the entire state-space.

Severe limitation, especially for problems with a large state-space.

Some nodes in the search can be *proven to be irrelevant to the outcome* of the search